

**Report for the Department of Climate
Change, Energy, the Environment and
Water (DCCEEW)
on
Reducing Microplastic Pollution in
Australia and Addressing its Impact on
the Country's Marine Ecosystem**

PROFESSIONAL PRACTICE IN POLICYMAKING (PPMN90010)

Word Count: **6,325.**

TABLE OF CONTENTS

EXECUTIVE SUMMARY	3
1. INTRODUCTION	5
2. BACKGROUND	6
2.1. Findings from the Literature Review	6
2.2. Current Legislative and Policy Landscape in Australia	7
2.3. International Comparisons and Case Studies	8
3. PROBLEM DEFINITION	10
3.1. Root Cause 1: Widespread Use of Microplastics in Consumer and Industrial Products	10
3.2. Root Cause 2: Insufficient Management of Wastewater Treatment Plants.....	11
3.3. Root Cause 3: Lack of Accountability for Microplastic Emissions from the Textile and Tyre Industries	12
4. POLICY ALTERNATIVES	14
4.1. Policy Option 1: Ban Intentionally Added Microplastics to Industrial and Consumer Products 14	
4.2. Policy Option 2: Mandatory Filtration Systems and Tertiary Treatments in Wastewater Treatment Plants (WWTPs).....	16
4.3. Policy Option 3: Extended Producer Responsibility (EPR) Programs for Synthetic Textile and Tyre Industries	17
5. MULTI-CRITERIA ANALYSIS	19
6. TRADE-OFFS AND RECOMMENDATION	23
7. CONCLUSION	26
8. REFERENCES	27
9. APPENDIX	34
i. Appendix A: Problem – Frame Stakeholder Map	34

EXECUTIVE SUMMARY

Microplastic pollution, similar to plastic pollution, is considered to be a wicked problem due to it being a complex and multifaceted issue with no single solution. This report aims to analyse the root causes and policy challenges of microplastic pollution in Australia and present three policy options to reduce its impact on the country's marine ecosystems, based on international and national precedence. The report is divided into 7 sections, starting with the introduction which gives a brief overview on the current and projected state of Australia's marine ecosystem and the impact of microplastic pollution on it. The next section provides an in-depth background to the overarching policy challenge and is divided into – (i) findings from the literature review, (ii) current legislative and political landscape in Australia and (iii) international comparisons and case studies. The report then introduces the three root causes: widespread use of microplastics in consumer and industrial products, insufficient management of wastewater treatment plants and lack of accountability for microplastic emissions from the textile and tyre industries. Based on the analysis so far in the report, the three policy alternatives are suggested:

- **Ban Intentionally Added Microplastics to Industrial and Consumer Products:** With precedence in the EU's REACH regulation, this option aims to directly target the use of microplastics in consumer goods, which often bypass wastewater treatment processes to enter marine ecosystems.
- **Mandatory Filtration Systems and Tertiary Treatments in Wastewater Treatment Plants (WWTPs):** This option addresses the inefficiencies of existing WWTP infrastructure, where microplastic particles are frequently discharged untreated into waterways.
- **Extended Producer Responsibility (EPR) Programs for Synthetic Textile and Tyre Industries:** This option addresses microplastic pollution in synthetic textile and tyre industries and aims to improve waste management by shifting the responsibility to manufacturers and holding them accountable for the environmental impact of their

products. Similar to Sweden's EPR programs, this policy encourages sustainable practices, such as improved materials and reducing microplastic production throughout the product's lifecycle.

A Multi-Criteria Analysis was employed in the next section to assess each policy option across the 5 criteria selected. Based on the MCA and the trade-offs and recommendations, the report finally suggests the third option, i.e., EPR programs targeted at synthetic textile and tyre industries to be the policy option recommended.

1. INTRODUCTION

In 2021, the United Nations Environment Programme's (UNEP) report on marine pollution projected that, in the absence of meaningful policy action, 'emissions from plastic waste into aquatic ecosystems' could triple by 2040. This means that global plastic waste entering our oceans could jump from 9-14 million tonnes per year in 2016 to an estimated 23-37 million tonnes per year by 2040 (United Nations Environment Programme, 2021, p. 3). Coral reefs across the Asia-Pacific are found to have 11.1 billion plastic items entangled on them, leading to outbreaks of disease in oceans (Lamb et al., 2018, p. 1). The Australia Institute (2024, p. 3) reports that plastic pollution represents 80% of marine litter, mostly in the form of microplastics, and kills an estimated 1 million seabirds and 100,000 sea mammals per year. Microplastics are produced when plastic degrades and dissolves into 'imperceptible smaller fragments' (The Australia Institute, 2024, p. 4) which the Australian Marine Conservation Society (2024, p.1) describes to be less than 5 millimetres in length. According to studies conducted across Australian waters (see Barrett et al., 2020; Senathirajah et al., 2020; Wootton et al., 2022), the amount of microplastics on the ocean floor is estimated at 14 million tonnes and has been found in the gastrointestinal tracts of 62% of fish in Australia, and in 94% of oysters globally. Despite this, Australian waste policy regarding the production of microplastics is vague and underdeveloped. Currently, the only production source of microplastics that is being targeted in federal policy is a voluntary scheme of phasing out microbeads (Australian Marine Conservation Society, 2024, p. 1). The Australian Capital Territory, Queensland, New South Wales, and Western Australia have all banned the supply of rinse-off personal care products containing microbeads (Accord, 2023, p.1). However, these bans currently only apply to personal care and cleaning products, leaving out other major sources of microbeads production such as household products and cosmetics. Furthermore, recycling was one of the cornerstones of the 2018 Australian National Waste Policy and the subsequent National Waste Plan. However, it has been found that substantial quantities of microplastics are generated during the process of plastic recycling (The Australia Institute, 2024, p. 18). This necessitates

a policy shift in how federal waste management policies should be shaped for Australia to successfully address the challenges of microplastic production and its impact on not only the country's marine ecosystem but also human health.

2. BACKGROUND

This section provides the necessary context required to understand the scope and extent of microplastic pollution in marine ecosystems, by examining both scientific and policy perspectives. In this section, the report explores the relevant findings from the literature review conducted and the current legislative and policy landscape in Australia for microplastic reduction. Finally, the last part of this section examines relevant international policies and case studies from which the policy alternatives (explored in section 5) are drawn from.

2.1. Findings from the Literature Review

Borriello & Rose (2021, p. 1) identify two types of microplastics which are of concern in relation to the environmental risk they pose to marine ecosystems. The first being primary microplastics, which are mainly found in personal care products, especially exfoliating scrubs and toothpastes in the form of microbeads. However, they can also be found in microfibre shedding from synthetic textiles such as clothing (Borriello & Rose, 2021, p. 1). The second type of microplastics which are a cause for environmental concern are secondary microplastics. These are the result of larger plastic pieces being broken down into smaller particles, often less than 5 millimetres (Australian Marine Conservation Society, 2024, p. 1; Borriello & Rose, 2021, p. 1). Studies conducted across marine ecosystems have shown that microplastic pollution is evident in most of the world's seas. According to Ashrafy et al. (2022, p. 4), microplastic pollution in Asia is mostly due to secondary microplastics while both European seas and its marine organisms have been reported to have trace elements of microplastic pollution. While Ashrafy et al. (2022, p. 4) suggests that higher microplastic concentrations in marine ecosystems could be due to a range of factors including tourism, urban expansion and environmental pressure, Freeman et al. (2020, p. 2) suggests four routes

through which both primary and secondary microplastics could enter marine environments. The routes suggested are: (i) emissions from primary microplastics during production, transportation and usage which enters through water streams, (ii) microplastics which were produced and weren't captured during wastewater treatments and enter via discharge into waterways, wear and tear of plastic components of wastewater treatment plants (WWTPs) and sewage sludge used in agricultural fertilisers, (iii) large plastic items disposed on land and eventually being washed up in marine ecosystems where they break down into secondary microplastics, and lastly (iv) disposed plastic items which break down into secondary microplastics on land and then enter marine ecosystems. This substantiates that marine ecosystems and, especially, the deep sea has become a sink for microplastic debris where it is consumed by marine organisms which inevitably enters into human food sources (Woodall et al., 2014, p. 7).

2.2. Current Legislative and Policy Landscape in Australia

Australia's legislative and policy framework concerning microplastic pollution can be looked at from the perspective of national strategies, state-specific initiatives, and alignment with international conventions. Unlike the European Union, where regulation has been in motion to reduce microplastic pollution (see European Commission, 2021; Halleux et al., 2023; The European Commission, 2023), Australian federal and state policies still lack direct microplastic-specific legislation. Current legislation and policies are mainly aimed at reducing general plastic waste or single-use plastic rather than targeted policies for microplastic reduction. In 2021, the Commonwealth Government announced its National Plastics Plan which briefly addressed the government's commitment to work towards phasing-in microfibre filters in new residential and commercial washings machine by 2030 (Department of Agriculture, Water and the Environment, 2021, p. 3). The Act also recognised microplastic pollution as a rising environmental concern and acknowledged wastewater as a key contributor to microplastics entering the country's marine ecosystem. However, other than a community stewardship program, there was no other policy mechanism mentioned for

managing microplastics in WWTTPs (Department of Agriculture, Water and the Environment, 2021, p. 9). Following the introduction of the *Recycling and Waste Reduction Act 2020*, microbeads were set to be phased-out in Australian rinse-off cosmetics, personal care and cleaning products. Furthermore, the introduction of the *Plastic Reduction and Circular Economy Act 2021* in New South Wales (NSW) further banned the supply of rinse-off personal care products which contained microbeads (NSW Government, 2002, p. 42). This was then followed by subsequent state governments in WA, ACT and QLD in 2023 (Accord, 2023, p. 1).

2.3. International Comparisons and Case Studies

Various international bodies have addressed the need for stricter policies in reducing microplastic pollution, including the United Nations, the World Health Organisation (WHO) and member states of ASEAN. The European Union has enacted various regulations and directives in order to reach its target goals of reducing microplastic pollution, such as issuing legislation to regulate microplastics under Annex XVII of the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH). Also known as Commission Regulation (EU) 2023/2025, the legislation was implemented in phases starting in 2023 to regulate synthetic polymer microplastics in products (Tsang, 2023, p. 2). One of its primary standards is to ensure that the percentage of these in products is kept below a specific threshold. Additionally, these microplastics are regulated based on their size, with the requirement being that at least 1% of particles (Tsang, 2023, p. 2) meet either of the following criteria:

- 5mm or smaller in all dimensions, or
- 15mm or smaller in length if the particle has a high length-to-width ratio, as commonly found in fibres with a length to diameter ratio greater than 3:1.

In the US, the Microbead-Free Waters Act was adopted in 2015 to ban the use of plastic microbeads in personal-care products, similar to Australia's own state-wise initiatives in banning microbeads from most household and personal-care products (McDevitt et al., 2017, p. 4). Meanwhile, the UN's incorporation of plastic pollution within the broader framework of the Sustainable Development Goals (SDGs), specifically under Target 14.1, emphasise the

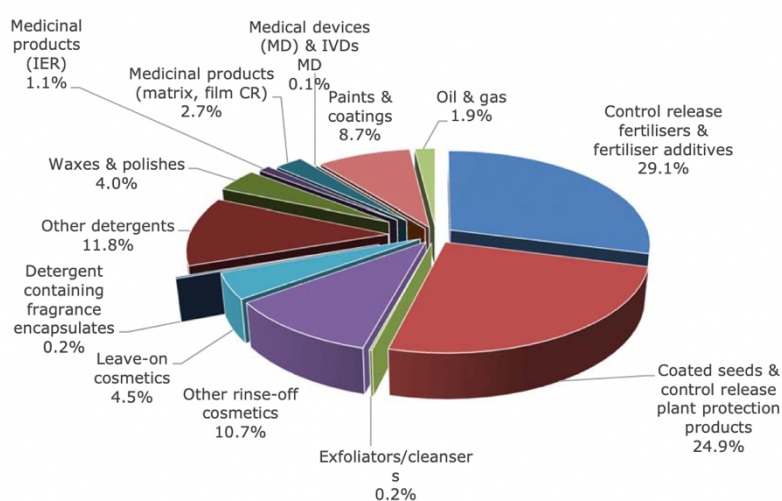
shift in global perception in reducing marine pollution (United Nations Sustainable Development, 2023, p. 1). Furthermore, in 2022, the state of California in the US introduced the Statewide Microplastics Strategy to manage microplastic pollution through source reduction and product stewardship (California Ocean Protection Council, 2022, p. 8). The Strategy employs a dual-track approach to combat microplastic pollution in California: (i) implementing immediate solutions that the state can undertake while furthering scientific understanding of microplastics and (ii) continuing to invest in and advance the scientific knowledge necessary to inform and refine future solutions (California Ocean Protection Council, 2022, p. 8).

3. PROBLEM DEFINITION

3.1. Root Cause 1: Widespread Use of Microplastics in Consumer and Industrial Products

Studies conducted across Australian freshwater systems have found the presence of microplastic polymers such as – polyethene and polypropylene – in sizes less than 2 millimetres in concentrations of 2.5803 g/L¹ and – polyethene, polyamide and polypropylene – of sizes less than 3 millimetres in concentrations of 0.216 g/L (Ashrafy et al., 2022, p. 4). These polymers are commonly found and used in various consumer and industrial products, such as polypropylene which is one of the most commonly used polymers in industries. It is found in medicine and in both the automotive as well as the packaging industry due to its high flexibility and durability (Craver & Carraher, 2000, p. 26). Furthermore, when urban wastewater networks gather water from these industries, they serve as a collection system for microplastics originating from various sources such as: (i) synthetic microfibres released during the manufacturing and wash-cycles of synthetic fabrics, (ii) plastic microbeads in rinse-off consumer products, and (iii) diffuse-source microplastics contained in urban runoff (OECD, 2020, p. 16).

Figure 1: Microplastic Use in Products as Share of Total Use (51,000 tonnes/annum)



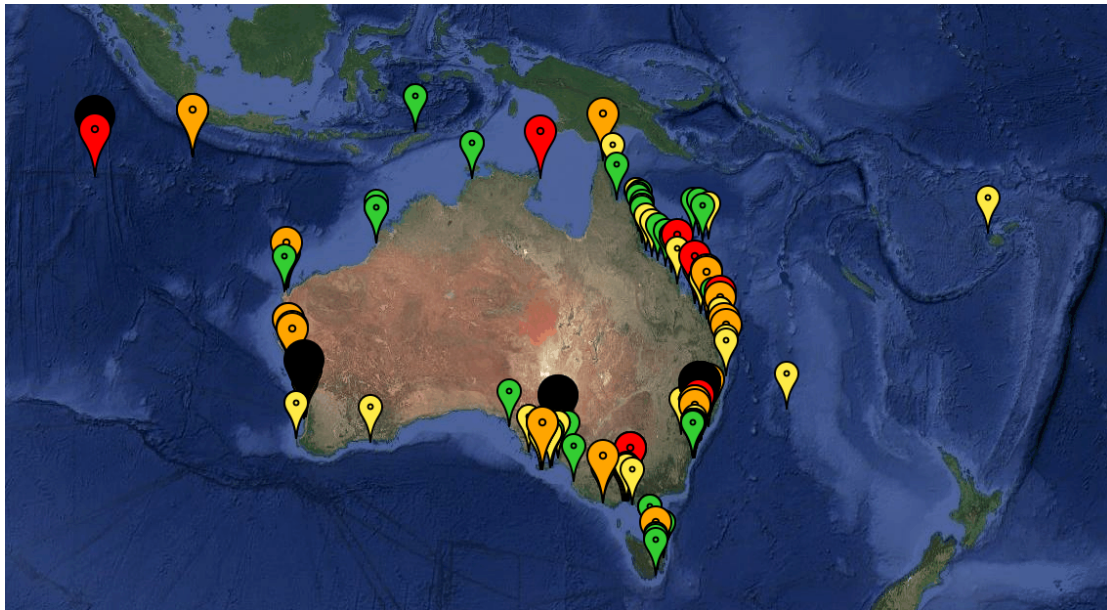
Source: (ECHA, 2019a, p. 78)

¹ g/L stands for grams per litter.

3.2. Root Cause 2: Insufficient Management of Wastewater Treatment Plants

Wastewater treatment plants (WWTPs) are recognized as a major route through which microplastics reach marine ecosystems. Due to their chemical properties, microplastics are highly resistant to both chemical and biological degradation and are primarily removed from WWTPs by binding to sludge during treatment. According to OECD (2020, p. 16), with over 80% of global wastewater released into marine ecosystems without treatment, urban wastewater acts as a significant pathway for microplastic pollution to increase. In OECD countries, of which Australia is a member, the situation is far better than its Sub-Saharan African and Asia-Pacific counterparts, with wastewater treatment technologies already in place. However, due to the large volumes of wastewater treated in Australia, substantial amounts of microplastics are still discharged from WWTPs into the country's marine ecosystem (Leusch & Ziajahromi, 2024, p. 1). Furthermore, Ashrafy et al. (2022, p. 2) indicates that approximately 17 microplastic polymers have been observed in urban wastewater which includes the likes of acrylate, high-density polyethylene, low-density polyethylene, melamine, Teflon and more unidentified polymers. Leusch & Ziajahromi (2024, p. 1) suggest that according to recent estimates, Australians release between 0.7g and 21g of microplastic per person into wastewater every year. Furthermore, according to two studies conducted by the Commonwealth Scientific and Industrial Research Organisation (CSIRO, 2020a; CSIRO, 2020b) on two WWTPs located in NSW, Malabar and Cronulla, it was found that: "Wastewater entering the Malabar and Cronulla treatment plants (WWTP) each day contained an estimated 24,000 – 61,000 million microplastic particles for Malabar, and 870 – 14,000 million for Cronulla (CSIRO, 2020b, p. 7). After treatment, the discharged water contained about 5,400 – 120,000 million microplastics at Malabar and 86 – 350 million at Cronulla." Furthermore, both the treatment plants showed similar types of microplastics, mainly polypropylene with smaller amounts of polyethylene, and polyethylene terephthalate (CSIRO, 2020b, p. 7).

Figure 1: Microplastics per square meter, found in the Australian Marine Ecosystem.



Source: (Hotspot Map | Ausmap, n.d.)

Note²: **Green** indicates 'very low' i.e., <10 mps/m², **Yellow** indicates 'low' i.e., between 11 – 50 mps/m², **Orange** indicates 'moderate' i.e., between 51 – 250 mps/m², **Red** indicates 'high' i.e., between 251 – 1,000 mps/m², **Black** indicates 'very high' i.e., between 1,001 – 10,000 mps/m² and **Purple** indicates 'extreme' i.e., > 10,000.

3.3. Root Cause 3: Lack of Accountability for Microplastic Emissions from the Textile and Tyre Industries

The textile and tire industries are major contributors of microplastic pollution. However, both these industries lack accountability for their emissions. The textile industry is a major contributor of microfibers released in marine ecosystems. According to Eriksen et al. (2018, p. 5), microplastics have several potential environmental release pathways but one of the biggest contributors of their passage through WTPPs to marine ecosystems is from the release of microfibres from textiles during wash-cycles. Similarly, tire wear particles significantly contribute to microplastic pollution and tyre wear microplastics (TWPs) are considered a major source of microplastics released in aquatic environments. Furthermore, studies indicate that washing synthetic clothes can release between 640,000 to 1.5 million microfibers per kilogram of fabric, contributing to an estimated 7.4 million metric tons of plastic pollution annually

²mps/m² stands for microplastics per square meter.

(Cotton Today, 2024, p. 1). These microfibers often pass through wastewater treatment plants (WWTPs) and enter aquatic environments, where they can harm marine life. According to Boucher et al. (2017, p. 5), approximately 35% of microplastics released into oceans globally originate from washing synthetic textiles. The European Environment Agency (2022, p. 2) estimates that synthetic textiles contribute between 0.2 and 0.5 million tonnes of microplastics in marine ecosystems each year. Furthermore, the abrasion of vehicle tyres during driving is a significant source of microplastic pollution, primarily through the release of TWP (European Environment Agency, 2022, p. 5). These particles consist of a mix of natural and synthetic rubber, with styrene-butadiene rubber typically accounting for approximately 60% of vehicle tyres (Khan et al., 2022, p. 356). As tyres wear down, they shed rubber and plastic fragments that can be washed off roads by wind or rain, ultimately ending up in roadside soils, wastewater, or directly in surface waters as microplastics (European Environment Agency, 2022, p. 5). Furthermore, current regulations often fail to hold manufacturers responsible for the environmental impact of their products throughout their lifecycle. This lack of accountability allows industries to continue practices that contribute to microplastic pollution without facing consequences.

4. POLICY ALTERNATIVES

This section of the report outlines three policy options for the Department of Climate Change, Energy, the Environment and Water (DCCEEW) to implement in order to reduce the acute impacts of microplastic pollution on the country's marine ecosystems.

4.1. Policy Option 1: Ban Intentionally Added Microplastics to Industrial and Consumer Products

This policy option suggests a federal ban that restricts microplastics intentionally added to industrial and consumer products, similar to the EU's ban under its REACH legislation (European Commission, 2023, p. 1) which has been more elaborately explored in Section 3. Intentionally added microplastics are considered to be tiny constituents of plastic which are specifically manufactured and added to products, and are often released into the environment during production or use due to the policy area being unregulated (European Commission, 2023, p. 1). According to the ECHA (2019, p. 96), which manages REACH, the ban would prevent the release of more than 500,000 tonnes of microplastics over the next 20 years in the EU and the European Economic Area (EEA) market. The proposal, which was outlined in the Annex XV Restriction Report, also includes restricting the concentration of microplastics in a mixture to 0.01% or less, banning products that release microplastics during use, and enforcing labelling requirements and safety data sheets to prevent microplastic release (ECHA, 2019b, p. 1). The federal ban this policy option suggests should include items such as loose plastic glitter, non-rinse off cosmetic products, agricultural products and paints where microplastics are often used as fillers (Plastic Soup Foundation, 2022, p. 5). Furthermore, in 2021, the Commonwealth Government passed the *Industrial Chemicals Environmental Management (Register) Act*, which established the IChEMS Register to classify industrial chemicals into seven schedules based on environmental risk (Department of Energy, Environment and Climate Action, 2024, p. 1). According to the Department of Climate Change, Energy, the Environment and Water (2024, p. 1), IChEMS establishes nationally consistent

standards to reduce the impact of industrial chemicals on the environment by managing and restricting its import, manufacture, export, use and disposal. For lower-risk chemicals under the Act, businesses are required to assess and share the environmental risks of the chemicals and 'apply practicable measures to eliminate these risks' (Department of Energy, Environment and Climate Action, 2024, p. 1). Industrial chemicals which are classified under higher-risk to the environment are placed under stricter control measures. The Act effectively places the responsibility for managing the introduction, use and disposal of industrial chemicals on industry players, including manufactures and importers (Department of Energy, Environment and Climate Action, 2024, p. 1) encouraging proactive risk management. To facilitate the implementation of this policy option, the DCCEEW should amend the Industrial Chemicals Environmental Management Standard (IChEMS) to specifically prohibit microplastics in consumer products by a defined deadline such as 2030. The legislation should further incorporate microplastic thresholds and require manufactures to provide evidence of compliance. This would have a direct environmental impact by preventing a significant source of primary microplastics, which often bypass WWTPs and accumulate in waterways (Ashrafy et al., 2022, p. 2), from entering marine ecosystems. Furthermore, aligning a federal ban on intentionally added microplastics to IChEMS is crucial due to the chemical properties of microplastics found in marine ecosystems, which is the focus of this policy report. According to research conducted by Ashrafy et al. (2022, p. 2), the most common types of microplastic polymer found in marine ecosystems are 'polyethylene (25%), polyethylene terephthalate (16.5%), polypropylene (14%), polyamide (12%), polystyrene (8.5%), polyvinyl alcohol (6%) and polyvinyl chloride (2%).'

4.2. Policy Option 2: Mandatory Filtration Systems and Tertiary Treatments in Wastewater Treatment Plants (WWTPs)

This policy option suggests that all municipal and industrial WWTPs should be required to install advanced filtration systems which capture microplastics before it is discharged into marine ecosystems. The policy estimates that enhanced filtration would act as a barrier to a major pathway of secondary microplastics from entering marine ecosystems. Given that microplastic particles are often too small for standard filters in existing wastewater systems, advanced filtration systems are crucial to reduce microplastic pollution. Case studies conducted by CSIRO across two WWTPs in NSW (see CSIRO, 2020a; CSIRO, 2020b) further highlight this by indicating the variance in efficiency of microplastic removal across treatment levels. For example, primary treatment at the Malabar WWTP achieves a removal rate of up to 79% but it primarily captures larger particles through sedimentation and screening (CSIRO, 2020b, p. 7). However, the treatment falls short when it comes to capturing smaller particles of microplastics. In stark contrast, tertiary treatments implemented at the Cronulla WWTP achieved removal rates as high as 98% by utilising advanced filtration and chemical treatments (CSIRO, 2020b, p. 3). Thereby, effectively capturing a broader range of microplastic sizes, including the smaller particles which often escape into marine environments post-treatment. By requiring WWTPs to incorporate advanced tertiary treatments or filtration systems could significantly reduce microplastic discharge via wastewater into marine ecosystems (Glen, 2024, p. 1). Furthermore, by adopting advanced treatment technologies and upgrading existing primary and secondary treatment processes to tertiary treatment systems such as Membrane Bioreactors (MBRs) or tertiary Sand-filters could further show higher removal efficiencies for microplastics compared to conventional treatment methods (Glen, 2024, p. 1). For example, MBRs use fine membranes that effectively filter out smaller plastic particles (Ameliex, 2024, p. 1). Treated wastewater is one of the biggest producers of microplastics globally, and removing residual microplastics from wastewater largely remains experimental, highlighting the need for further innovation as well. To advance

these technologies, policy support and regulatory incentives would be crucial to encourage industry stakeholders to develop more effective filtration and tertiary treatment methods.

4.3. Policy Option 3: Extended Producer Responsibility (EPR) Programs for Synthetic Textile and Tyre Industries

Secondary microplastics such as synthetic microfibres, tyre wear particles, paint flakes and other land-based microplastics cause significant damage to marine ecosystems as they often release unintentional emissions occurring during the usage phase, either due to abrasions or wear and tear (OECD, 2020, p. 15). The third policy option suggests introducing Extended Producer Responsibility (EPR) programs specifically targeting synthetic textile and tyre industries, which are known to release some of the highest microplastic emissions during its usage cycle. EPR is a tested policy framework which specialises in reducing risks associated with waste management. EPR programs require producers to be both legally and financially responsible for mitigating the environmental risks associated with their products' lifecycle stages (Nikiema & Asiedu, 2022, p. 24552). According to Eriksen et al. (2018, p. 12), EPR programs can further limit the health, safety, environmental and social impacts of plastic products through prevention and mitigation strategies once enforced. In 2013, California's National Resources Defence Council introduced an EPR program aimed at holding producers accountable not only for recycling but also for mitigating and preventing litter. Under the program, producers were asked to reduce their products' total volume in the environment by 75% and 95% in six years and eleven years, respectively (Nikiema & Asiedu, 2022, p. 24552; Eriksen et al., 2018, p.13). This can be applied within the EPR programs for synthetic textile and tyre industries, requiring manufacturers to take responsibility of not only recycling their products but also reducing the emissions from the microplastics released from their products. Similar to California's policy initiatives, textile and tyre producers could be mandated to decrease their environmental volume of microplastic emissions by specific targets—such as 75% within six years and 95% within eleven years. This would encourage industries to innovate in product design to minimize microplastic shedding and adopt sustainable materials.

The policy option also further recommends amending the *Product Stewardship Act 2011* to include synthetic textiles and tyre manufacturers. According to the Australian Competition and Consumer Commission (2024, p. 1), the existing Tyre Product Stewardship Scheme has been authorised to continue for 3 more years. The report recommends that the Scheme could incorporate microplastic reduction goals by prioritising R&D, improved product and material design and utilising end-of-life recycling initiatives to reduce the release of microplastic during its usage cycle without compromising tyre performance. The report also suggests investing in further R&D in analysing the environmental impacts of tyre wear particles (TWPs) which are a significant source of microplastic emissions from the tyre industry.

5. MULTI-CRITERIA ANALYSIS

This section of the report evaluates the effectiveness of the three policy alternatives by conducting a multi-criteria analysis (MCA). According to Kraft and Furlong (2023, p. 166), evaluative criteria are the 'specific dimensions of policy objectives' which are used to assess and compare policy options or existing policies. They can be used to provide clear standards in order to ensure that analysis of a policy is objective and focused on the issues that are important to the client. Furthermore, evaluative criteria are also essential in ranking policy alternatives in order of preference (Kraft & Furlong, 2023, p. 166). Selecting criteria that aligns with the policy objectives, therefore, is crucial. Kraft and Furlong (2023, p. 166) highlight this in their examples of how some criteria are better suited for issues such as 'access to healthcare' than agricultural subsidies or educational reforms. Therefore, using the right evaluative criteria ensures that the analysis targets key concerns important to the client and makes policy evaluation more structured and effective. The following five evaluative criteria were chosen for the analysis, with each criterion assigned a weightage according to its importance in reducing microplastic pollution:

- **Environmental Impact** evaluates each policy option's direct and indirect effects on the environment and determines its overall contribution to the health of marine ecosystems. As the primary goal of this report is to reduce microplastic pollution, this criterion is assigned the highest weightage at 30%.
- **Cost Effectiveness** assesses the upfront and operational costs of implementing a policy as well as the long-term benefits or liabilities for different stakeholders – such as WTP operators, manufacturers, and regulatory bodies. It also considers the cost of replacement materials and infrastructure upgrades required. The weight assigned to this criterion is 20% as evaluating cost-effectiveness is crucial to gain both government and industry support.
- **Social Acceptability** defines the extent to which the public might accept or support a policy option (Kraft & Furlong, 2023, p. 166). Furthermore, social acceptability is

essential for policies which involve consumer goods and/or behavioural shifts, as seen in most environmental policies. However, the weight assigned to it is 15%, which is lower than environmental impact, cost-effectiveness and technical feasibility as the policy challenge requires the other three criteria to be prioritised.

- **Stakeholder Perception** encompasses the outlooks of the various stakeholders invested in the policy option which in environmental policies could include manufacturers, fossil fuel companies, the government and environmental advocacy groups. This criterion, therefore, assesses the willingness of these stakeholders to support the policy option. It is weighted similarly to 'social acceptability' at 15% because while 'stakeholder perception' can drive investment and support particularly from industry operators, it should not overshadow the importance of the environmental benefits of the policy option.
- **Technical Feasibility** is the availability as well as the reliability of the technology that is required in order for the policy to be implemented (Kraft & Furlong, 2023, p. 166). The weightage assigned to it is similar to 'cost-effectiveness' at 20% due to the importance of this criterion, especially in environmental policies where technological limitations can seriously hurt the prospects of a policy option.

The weightage assigned to the criteria prioritises environmental impact, cost effectiveness and technical feasibility due to its relative importance over social acceptability and stakeholder perception in producing the most sustainable outcomes. Therefore, the goal of the five criteria proposed is to provide a balanced evaluation of the key dimensions explored in the policy options, i.e., environmental, economic, social and technical.

Table 1: Multi-criteria analysis (MCA) of the three policy alternatives

Note: A detailed problem-frame stakeholder map could be found in Appendix A.

Assessment Criteria: Reducing Microplastic Pollution in Australian Marine Ecosystems						
Weight (%)	30%	20%	15%	15%	20%	
Policy Alternatives	Environmental Impact	Cost Effectiveness	Social Acceptability	Stakeholder Perception	Technical Feasibility	Weight
	0	0	0	0	0	
Status Quo: No Reduction in Microplastic Pollution						
Policy Option 1: Ban Intentionally Added Microplastics to Industrial and Consumer Products	High: 3 Banning intentionally added microplastics would significantly reduce direct pollution sources by preventing their entry into marine environments.	Lower Medium: 1 Initial costs might be moderate for manufactures who would need to use alternative materials for their products. However, long-term benefits include reduced environmental cleanup and healthcare costs.	Upper Medium: 2 Moderate public support is expected as awareness around microplastics is growing. Consumers are also increasingly seeking eco-friendly products. However, there might be conservative consumers who might be opposed.	Lower Medium: 1 Consumer goods industries may resist due to the additional costs of alternative materials. However, eco-conscious brands are likely to support the change and see it as an opportunity to enhance their corporate image.	Upper Medium: 2 The ban is feasible technically, as alternate materials exist. However, product reformulation may require R&D investment.	0.0195 2
Policy Option 2: Mandatory Filtration Systems and Tertiary Treatments in Wastewater Treatment Plants (WWTPs)	High: 3 Would be effective in capturing a wide range of microplastics before they reach marine environments – e.g., tertiary sand filters can provide an additional 79% microplastic retention compared to existing secondary treatments in WWTPS	Low: -1 High capital and operational costs are required for advanced filtration systems, which may strain public utilities and lead to higher water treatment fees.	Lower Medium: 1 Mixed public support as there may be concerns about potential increases in price for water services, even though the environmental benefits are recognised.	Low: -1 Wastewater operators may see this as costly and challenging without financial incentives, but environmental advocates would likely support it.	Lower Medium: 1 Technically feasible but challenging as high filtration requirements demand energy-intensive technologies, impacting the carbon footprint of treatment plants; plus, longer timeframe to be deployed (Freeman et al., 2020, p. 7).	0.009 3

	(Funck et al., 2021, p. 1).						
Policy Option 3: Extended Producer Responsibility (EPR) Programs for Synthetic Textile and Tyre Industries	High: 3 Reduces microplastic emissions from two major sources (textiles and tires). Manufacturers would be required to adopt sustainable materials and recycling initiatives.	Lower Medium: 1 Shifts costs from municipalities to producers, potentially reducing public spending on cleanup while encouraging sustainable design.	High: 3 High social acceptability as public perception in the past few years have shifted to support sustainable initiatives, and preferences for companies with eco-conscious practices are rising.	Upper Medium: 2 Positive support from environmental NGOs and sustainability-focused investors. However, traditional industries may resist due to increased operational costs.	Upper Medium: 2 Feasible, with existing EPR programs providing a framework. However, scaling may require industry adaptation to new standards and recycling technologies.	0.0225	1

While conducting the MCA, a comparative score was assigned to each policy option. This was to prioritise the option which would be the most effective, feasible and acceptable based on the challenges of reducing microplastic pollution in Australia. As can be seen from the table above, policy option 3 was chosen as the preferred policy to recommend.

6. TRADE-OFFS AND RECOMMENDATION

This policy report presented three policy options to be considered in order to address the root causes introduced in section 4. This section of the report now analyses the existing trade-offs between the five evaluative criteria against which the three policy options were weighted, and recommends policy option 3 as the most suitable policy out of the three to be implemented. According to Huang et al. (2011, p. 3578), decision-making in environmental policy often requires serious consideration in terms of the trade-offs that exist between economic, socio-political and environmental impacts, which is then further complicated because of multiple stakeholder interests. The base option, or status quo as referred to in the table, maintains that there is no reduction in microplastic pollution.

The first policy option: banning intentionally-added microplastics to products – specifically focuses on banning microplastics in consumer and industrial products, therefore, encouraging manufacturers to switch to more sustainable and alternate materials. Research has shown that environmental regulation when initiated with subsidies for innovation incentivises investment in research and development (Freeman et al., 2020, p. 7). This leads to the creation of clearer industry standards in refining existing technologies as well as innovation in new technologies which would comply with the introduced regulation (Freeman et al., 2020, p. 7). This is particularly of importance for environmental regulation where progress in sustainable technologies is crucial for future policy effectiveness. This option scores high in ‘environmental impact’ due to the policy option’s strategy in directly reducing microplastics at one of its primary sources. The option scores medium across the four other criteria due to the (i) high costs related to transitioning to substitute materials, (ii) positive social acceptability due to growing awareness of microplastics globally but also resistance from traditional industry associations, (iii) mixed feedback from stakeholders – with environmental groups being in favour but potentially being seen as too restrictive by producers, and finally (iv) being feasible in terms of enforcing legislation but the lack of cost-effective substitutes makes it challenging to fully implement.

The second policy option: mandatory filtration systems in wastewater treatment plants (WWTPs) – addresses microplastic pollution during the process of treating wastewater and its byproduct, sludge. This option scored high for ‘environmental impact’ due to its perceived effectiveness in capturing microplastics at a critical point before they reach marine ecosystems. It scored ‘low’ in cost effectiveness due to the high costs associated with the installation, operation and maintenance of advanced filtration technologies. Additionally, this point co-relates to the ‘lower medium’ score assigned to it for ‘social acceptability’ as these high costs could eventually be passed on to consumers or local governments, further raising water utility costs and straining municipal budgets. Stakeholder perception for this policy option scores ‘low’ due to the high energy costs imposed on WWTP operators. And ‘technical feasibility’ scores ‘lower medium’ due to additional processing requirements and operational complexities. According to Freeman et al. (2020, p. 7), the wastewater treatment industry is still highly conservative in nature and combined with a tough financial market and existing regulatory structures already in place, any new technology could take a minimum of 5 years to be implemented. Furthermore, the energy requirement for wastewater treatment in Australia is estimated to be at 1000 gigawatt hours (GWh)² a year (NSW Government & Office of Environment and Heritage, 2019, p. 3). This is equated to be the same amount of power consumed by 170,000 households in New South Wales and 0.4% of Australian electricity consumption (NSW Government & Office of Environment and Heritage, 2019, p. 3).

The third policy option: extended producer responsibility (EPR) programs for synthetic textiles and tyres – was chosen to be the preferred policy option during the MCA analysis to be recommended out of the three alternatives. EPR programs generally score high on ‘environmental impact’ as they transfer the environmental costs of microplastic pollution back to manufacturers (OECD, 2024, p. 10). Therefore, they further incentivise industries to adopt alternate materials and eco-friendly practices for production. Unlike the second option, where consumers and municipalities could eventually bear the high costs of implementation, EPR shifts the financial and environmental responsibility from municipalities and consumers to

producers (OECD, 2024, p. 10). Therefore, the policy option scores high both on ‘environmental impact’ and ‘social acceptability’.

Based on the MCA conducted, the third policy option, i.e., implementing EPR programs for synthetic textiles and tyres, is the recommended policy choice this report would suggest. Each option involved trade-offs between the 5 evaluative criteria selected for the analysis. For instance, Policy Option 1 provides strong environmental benefits but scores low in cost-effectiveness and technical feasibility due to high costs and lack of R&D in alternate materials. This further leads to a trade-off between environmental impact and cost-effectiveness as industries may find it difficult to comply without incurring significant costs. While Policy Option 2 ranked high in environmental impact as well, it ranked lower in cost-effectiveness and technical feasibility due to the high energy costs and complex infrastructure requirements needed. Therefore, showcasing a trade-off between immediate environmental gains versus long-term financial viability. Finally, Policy Option 3 had the most balanced trade-offs across all 5 criteria, having a high environmental impact while distributing costs across the supply chain and shifting the responsibility to producers – leading to higher social acceptability than the other two options. Furthermore, this option minimises the financial impact on any one group and leverages the existing EPR regulatory framework in Australia.

CONCLUSION

This report recommends the implementation of the third policy option, i.e., implementing EPR programs for synthetic textile and tyre industries, as the most effective policy solution to address microplastic pollution. Through a multi-criteria analysis (MCA), EPR scored highest for balancing environmental impact, social acceptability, cost-effectiveness, and technical feasibility. Unlike other policy options, EPR programs shift the responsibility for microplastic emissions to producers, incentivising sustainable practices and innovation while reducing the financial strain on consumers and municipalities. The MCA highlighted EPR's advantages in effectively aligning with environmental impact and social acceptability, as it promotes industry accountability without imposing undue burdens on consumers. By implementing EPR programs as a nationwide strategy, Australia can position itself as a leader in combating microplastic pollution in its marine ecosystems. Furthermore, to enhance the effectiveness of EPR programs, the report also recommends (i) further research on consumer behaviour towards sustainable products to evaluate the long-term impact of EPR on microplastic reduction, and (ii) establishing a robust framework for monitoring and evaluating the effectiveness of EPR programs over time.

REFERENCES

- Accord. (2023). *Coordinating Australia's voluntary industry phase-out of plastic microbeads in rinse-off cosmetics, personal care and cleaning products*. <https://accord.asn.au/sustainability/beadrecede/>
- Ameliex. (2024, June 17). *Membrane Bioreactors (MBR) for wastewater treatment*. PCI Membranes. <https://www.pcimembranes.com/articles/membrane-bioreactors-mbr-for-wastewater-treatment/>
- Ashrafy, A., Liza, A. A., Islam, M. N., Billah, M. M., Arafat, S. T., Rahman, M. M., & Rahman, S. M. (2022). Microplastics Pollution: A brief review of its source and abundance in different aquatic ecosystems. *Journal of Hazardous Materials Advances*, 9, 100215. <https://doi.org/10.1016/j.hazadv.2022.100215>
- Australian Competition and Consumer Commission. (2024, September 2). *ACCC reauthorises Tyre Product Stewardship Scheme*. <https://www.accc.gov.au/about-us/news/media-updates/accc-reauthorises-tyre-product-stewardship-scheme>
- Australian Marine Conservation Society. (2024, August 30). *Microplastics: tiny plastics, big problem*. <https://www.marineconservation.org.au/microplastics/>
- Barrett, J., Chase, Z., Zhang, J., Holl, M. M. B., Willis, K., Williams, A., Hardesty, B. D., & Wilcox, C. (2020). Microplastic pollution in Deep-Sea sediments from the Great Australian Bight. *Frontiers in Marine Science*, 7. <https://doi.org/10.3389/fmars.2020.576170>
- Borriello, A. (2023). Preferences for microplastic marine pollution management strategies: An analysis of barriers and enablers for more sustainable choices. *Journal of Environmental Management*, 344, 118382. <https://doi.org/10.1016/j.jenvman.2023.118382>
- Borriello, A., & Rose, J. M. (2021). The issue of microplastic in the oceans: Preferences and willingness to pay to tackle the issue in Australia. *Marine Policy*, 135, 104875. <https://doi.org/10.1016/j.marpol.2021.104875>
- Boucher, J., Friot, D., & International Union for Conservation of Nature. (2017). Primary Microplastics in the Oceans: A Global Evaluation of sources. In C. G. Lundin & J. Matos De Sousa (Eds.), *IUCN* (p. 43). IUCN, Gland,

- Switzerland. <https://portals.iucn.org/library/sites/library/files/documents/2017-002-En.pdf>
- California Ocean Protection Council. (2022). *Statewide Microplastics Strategy Understanding and Addressing Impacts to Protect Coastal and Ocean Health*. https://www.opc.ca.gov/webmaster/ftp/pdf/agenda_items/20220223/Item_6_Exhibit_A_Statewide_Microplastics_Strategy.pdf
- Clausen, L. P. W., Hansen, O. F. H., Oturai, N. B., Syberg, K., & Hansen, S. F. (2020). Stakeholder analysis with regard to a recent European restriction proposal on microplastics. *PLoS ONE*, 15(6), e0235062. <https://doi.org/10.1371/journal.pone.0235062>
- Cotton Today. (2024, October 14). *The urgency of addressing plastic pollution in the apparel industry*. CottonToday. <https://cottontoday.cottoninc.com/urgency-of-addressing-plastic-pollution-in-apparel-industry/#:~:text=Our%20research%20reveals%20that%20synthetic,end%20of%20their%20life%20cycle.&text=This%20compound%20the%20well%2Ddocumented,during%20laundrying%20and%20daily%20wear>.
- Craver, C. D., & Carraher, C. E. (2000). INTRODUCTION TO POLYMER SCIENCE AND TECHNOLOGY. In *Elsevier eBooks*(pp. 21–47). <https://doi.org/10.1016/b978-008043417-9/50004-0>
- CSIRO. (2020a). Microplastic quantification in wastewater: Wastewater influent and effluent trends over a 10 month period. *CSIRO*. <https://www.epa.nsw.gov.au/-/media/epa/corporate-site/resources/waste/csiro-report-microplastic-quantification-in-wastewater.pdf>
- CSIRO. (2020b). Microplastics in wastewater. *CSIRO*. <https://www.epa.nsw.gov.au/-/media/epa/corporate-site/resources/waste/csiro-report-microplastics-in-wastewater.pdf>
- Department of Agriculture, Water and the Environment. (2021). *National Plastics Plan 2021*. <https://www.agriculture.gov.au/sites/default/files/documents/national-plastics-plan-2021.pdf>
- Department of Climate Change, Energy, the Environment and Water. (n.d.). *Industrial Chemicals Environmental Management Standard -*

- IChEMS*. <https://www.dcceew.gov.au/environment/protection/chemicals-management/national-standard>
- Department of Climate Change, Energy, the Environment and Water. (2018). *Final Report: Assessment of the voluntary phase out of Microbeads*. <https://www.dcceew.gov.au/sites/default/files/documents/assessment-voluntary-phase-out-microbeads.pdf>
- Department of Energy, Environment and Climate Action. (2024). *Industrial Chemicals Environmental Management Standard (ICHEMS)*. Victorian Department of Energy, Environment and Climate Action. <https://www.environment.vic.gov.au/sustainability/industrial-chemicals-environmental-management-standard>
- ECHA. (2019a). ANNEX XV Restriction Report: Proposal for a Restriction - intentionally added microplastics. In *ECHA*. <https://echa.europa.eu/documents/10162/05bd96e3-b969-0a7c-c6d0-441182893720>
- ECHA. (2019b). ECHA proposes to restrict intentionally added microplastics. In *ECHA*. <https://echa.europa.eu/documents/10162/b949137e-57a6-dde0-9f62-7ad0f58b123b#:~:text=The%20Annex%20XV%20report%20outlines,consistent%20with%20a%20'microplastic'>.
- Eriksen, M., Thiel, M., Prindiville, M., & Kiessling, T. (2018). Microplastic: What are the solutions? In *Wagner, M., Lambert, S. (eds) Freshwater Microplastics. The Handbook of Environmental Chemistry* (pp. 273–298). https://doi.org/10.1007/978-3-319-61615-5_13
- European Commission. (2021). *Pathway to a Healthy Planet for all: EU Action Plan: “Towards Zero Pollution for Air, Water and Soil.”* <https://faolex.fao.org/docs/pdf/eur204204.pdf>
- European Commission. (2023). Protecting environment and health: Commission adopts measures to restrict intentionally added microplastics. In *European Commission - Press Release*.
- European Environment Agency. (2022). Microplastic pollution from textile consumption in Europe. In *Eionet Report - ETC/CE 2022/1*. <https://www.eionet.europa.eu/etcs/etc->

- ce/products/etc-ce-products/etc-ce-report-1-2022-microplastic-pollution-from-textile-consumption-in-europe/@@download/file/ETC_microplastics%20and%20textiles.pdf
- Freeman, S., Booth, A. M., Sabbah, I., Tiller, R., Dierking, J., Klun, K., Rotter, A., Ben-David, E., Javidpour, J., & Angel, D. L. (2020). Between source and sea: The role of wastewater treatment in reducing marine microplastics. *Journal of Environmental Management*, 266, 110642. <https://doi.org/10.1016/j.jenvman.2020.110642>
- Funck, M., Al-Azzawi, M. S., Yildirim, A., Knoop, O., Schmidt, T. C., Drewes, J. E., & Tuerk, J. (2021). Release of microplastic particles to the aquatic environment via wastewater treatment plants: The impact of sand filters as tertiary treatment. *Chemical Engineering Journal*, 426, 130933. <https://doi.org/10.1016/j.cej.2021.130933>
- Glen, R. (2024, May 6). *Turning back the microplastic tide*. True Water. <https://truewater.com.au/microplastics-in-sewage/>
- Halleux, V., João Albuquerque, Catherine Chabaud, Ska Keller, Karol Karski, & Idoia Villanueva Ruiz. (2023). Reducing Microplastic Pollution From Plastic Pellet Losses. In *European Parliament*. European Parliamentary Research Service. [https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/760442/EPRS_BRI\(2024\)760442_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/760442/EPRS_BRI(2024)760442_EN.pdf)
- Hossain, M. S. (2024). People's attitudes regarding plastics and microplastics pollution: Perceptions, behaviors, and policy implications. *Marine Policy*, 165, 106219. <https://doi.org/10.1016/j.marpol.2024.106219>
- Hotspot Map | ausmap*. (n.d.). Ausmap. <https://www.ausmap.org/hotspot-map>
- Huang, I. B., Keisler, J., & Linkov, I. (2011). Multi-criteria decision analysis in environmental sciences: Ten years of applications and trends. *The Science of the Total Environment*, 409(19), 3578–3594. <https://doi.org/10.1016/j.scitotenv.2011.06.022>
- Khan, A., Kian, L. K., Jawaid, M., Khan, A. a. P., Marwani, H. M., Alotaibi, M. M., & Asiri, A. M. (2022). Preparation and characterization of lignin/nano graphene oxide/styrene butadiene rubber composite for automobile tyre application. *International Journal of Biological Macromolecules*, 206, 363–370. <https://doi.org/10.1016/j.ijbiomac.2022.02.146>

- Komyakova, V., Vince, J., & Haward, M. (2020). Microplastics and the Australian marine environment: issues and options. In National Environmental Science Program, Marine Biodiversity Hub, *Report to the National Environmental Science Program, Marine Biodiversity Hub*. IMAS, University of Tasmania. <https://creativecommons.org/licenses/by/4.0/>
- Kraft, M. E., & Furlong, S. R. (2023). *Public policy: Politics, Analysis, and Alternatives*. CQ Press.
- Lamb, J. B., Willis, B. L., Fiorenza, E. A., Couch, C. S., Howard, R., Rader, D. N., True, J. D., Kelly, L. A., Ahmad, A., Jompa, J., & Harvell, C. D. (2018). Plastic waste associated with disease on coral reefs. *Science*, 359(6374), 460–462. <https://doi.org/10.1126/science.aar3320>
- Leusch, F., & Ziajahromi, S. (2024). *Australians are washing microplastics down the drain and it's ending up on our farms*. The Conversation. <https://theconversation.com/australians-are-washing-microplastics-down-the-drain-and-its-ending-up-on-our-farms-223079>
- May, P. J. (2005). Policy Maps and Political Feasibility. In *Thinking Like a Policy Analyst* (pp. 127–151). Palgrave Macmillan, New York. https://doi.org/10.1057/9781403980939_7
- McDevitt, J. P., Criddle, C. S., Morse, M., Hale, R. C., Bott, C. B., & Rochman, C. M. (2017). Addressing the issue of microplastics in the wake of the Microbead-Free Waters Act—A new standard can facilitate improved policy. *Environmental Science & Technology*, 51(12), 6611–6617. <https://doi.org/10.1021/acs.est.6b05812>
- Nikiema, J., & Asiedu, Z. (2022). A review of the cost and effectiveness of solutions to address plastic pollution. *Environmental Science and Pollution Research*, 29(17), 24547–24573. <https://doi.org/10.1007/s11356-021-18038-5>
- NSW Government. (2002). *Plastic Reduction and Circular Economy Act 2021 No. 31*. <https://legislation.nsw.gov.au/view/whole/html/inforce/current/act-2021-031>
- NSW Government & Office of Environment and Heritage. (2019). *Energy Efficiency Opportunities in Wastewater Treatment Facilities* [Report]. Office of Environment & Heritage. <https://www.environment.nsw.gov.au/resources/business/wastewater-treatment-facilities-energy-efficiency-opportunities-190114.pdf>

- OECD. (2020). Policies to Reduce Microplastics in Water: Focus on Textiles and Tyres. In *OECD*. [https://one.oecd.org/document/ENV/EPOC/WPRPW\(2020\)3/FINAL/en/pdf](https://one.oecd.org/document/ENV/EPOC/WPRPW(2020)3/FINAL/en/pdf)
- OECD. (2024). Extended Producer Responsibility: Basic facts and key principles. In *OECD Environment Policy Papers, No. 41*. OECD Publishing, Paris. <https://doi.org/10.1787/67587b0b-en>
- Office of Parliamentary Counsel. (2020). *Recycling and Waste Reduction Act 2020*. <https://www.legislation.gov.au/C2020A00119/latest/text>
- Plastic Soup Foundation. (2022). *PLASTIC THE HIDDEN BEAUTY INGREDIENT: An analysis of the use of microplastics in personal care products and the upcoming legislation covering intentionally added microplastics*. <https://www.beatthemicrobead.org/wp-content/uploads/2022/06/Plastic-TheHiddenBeautyIngredients.pdf>
- Senathirajah, K., Attwood, S., Bhagwat, G., Carbery, M., Wilson, S., & Palanisami, T. (2020). Estimation of the mass of microplastics ingested – A pivotal first step towards human health risk assessment. *Journal of Hazardous Materials*, 404, 124004. <https://doi.org/10.1016/j.jhazmat.2020.124004>
- The Australia Institute. (2024). Plastic waste in Australia. In *Discussion Paper [Report]*. <https://australiainstitute.org.au/wp-content/uploads/2024/01/P1482-Plastic-Waste-in-Australia-Web.pdf>
- The European Commission. (2023). *Commission Regulation (EU) 2023/2055*. Official Journal of the European Union. <http://data.europa.eu/eli/reg/2023/2055/oj>
- Tsang, H. (2023, October 2). *EU Regulates Microplastics under REACH*. SGS Corp. <https://www.sgs.com/en/news/2023/10/safeguards-12623-eu-regulates-microplastics-under-reach>
- United Nations Environment Programme. (2021). *From Pollution to Solution: A Global Assessment of Marine litter and Plastic pollution*. <https://wedocs.unep.org/bitstream/handle/20.500.11822/36965/POLSOLSu m.pdf?sequence=24&isAllowed=y>
- United Nations Sustainable Development. (2023, October 20). *Oceans - United Nations Sustainable Development*. <https://www.un.org/sustainabledevelopment/oceans/#:~:text=14.1%20>

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- Usman, S., Razis, A. F. A., Shaari, K., Azmai, M. N. A., Saad, M. Z., Isa, N. M., & Nazarudin, M. F. (2022). The burden of microplastics pollution and contending policies and regulations. *International Journal of Environmental Research and Public Health*, 19(11), 6773. <https://doi.org/10.3390/ijerph19116773>
- Woodall, L. C., Sanchez-Vidal, A., Canals, M., Paterson, G. L., Coppock, R., Sleight, V., Calafat, A., Rogers, A. D., Narayanaswamy, B. E., & Thompson, R. C. (2014). The deep sea is a major sink for microplastic debris. *Royal Society Open Science*, 1(4), 140317. <https://doi.org/10.1098/rsos.140317>
- Wootton, N., Sarakinis, K., Varea, R., Reis-Santos, P., & Gillanders, B. M. (2022). Microplastic in oysters: A review of global trends and comparison to southern Australia. *Chemosphere*, 307, 136065. <https://doi.org/10.1016/j.chemosphere.2022.136065>
- Ziajahromi, S., Neale, P. A., Rintoul, L., & Leusch, F. D. (2017). Wastewater treatment plants as a pathway for microplastics: Development of a new approach to sample wastewater-based microplastics. *Water Research*, 112, 93–99. <https://doi.org/10.1016/j.watres.2017.01.042>

APPENDIX

Appendix A: Problem – Frame Stakeholder Map

